№6

2026-05-02

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   - Quarto.
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   - jupyter notebook;
   - Quarto.
10. jupyter notebook .

4

4.

4.

: - S ( ) — ; - I ( ) —
; - R (/) — .
: - infection: $S + I \to I + I$ (β); - recovery: $I \to R$ (γ)

5

5.

5.1

DrWatson julia (. 1).

```
julia> Pkg.add("DrWatson")
    Updating registry at `~/.julia/registries/General.toml`
┌ Error: Some registries failed to update:
│     — /home/mwakutaipa/.julia/registries/General.toml — failed to d
ownload from https://pkg.julialang.org/registry/23338594-aafe-5451-b9
3e-139f81909106/31c1bca344f440e166a96c5b4dca8b6c99b1b120. Exception:
RequestError: HTTP/2 200 (Recv failure: Connection reset by peer) whi
le requesting https://pkg.julialang.org/registry/23338594-aafe-5451-b
93e-139f81909106/31c1bca344f440e166a96c5b4dca8b6c99b1b120
└ @ Pkg.Registry ~/.julia/juliaup/julia-1.10.10+0.x64.linux.gnu/share
/julia/stdlib/v1.10/Pkg/src/Registry/Registry.jl:528
   Resolving package versions...
  No Changes to `~/.julia/environments/v1.10/Project.toml`
  No Changes to `~/.julia/environments/v1.10/Manifest.toml`

julia> using DrWatson

julia> initialize_project("project"; authors="Mildred", git=false)
```

1:

(. 2)

```
julia> cd("project")

julia> Pkg.add("AlgebraicPetri")
   Resolving package versions...
   Installed XML2_jll ──────────── v2.15.1+0
   Installed NameResolution ─────── v0.1.5
   Installed LabelledArrays ─────── v1.19.0
   Installed GeneralizedGenerated — v0.3.3
 Progress [========>                                ]  4/22
```

2:

5.3

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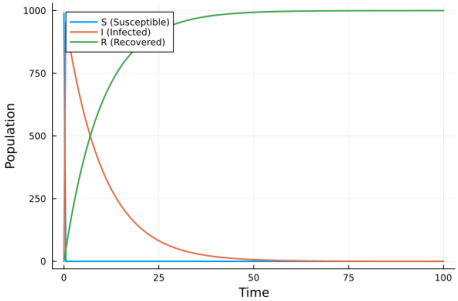
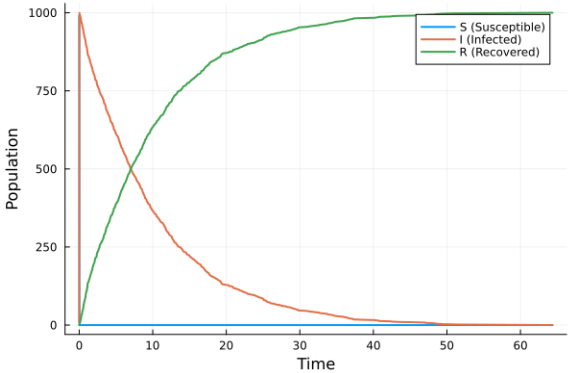
5.4 SIR

. beta gamma,
: (. 3) (. 4).
CSV



3: SIR

5.5 SIR


S (Susceptible)
I (Infected)
R (Recovered)
Population
Time
1000
750
500
250
0
0
10
20
30
40
50
60

4: SIR

5.6

.

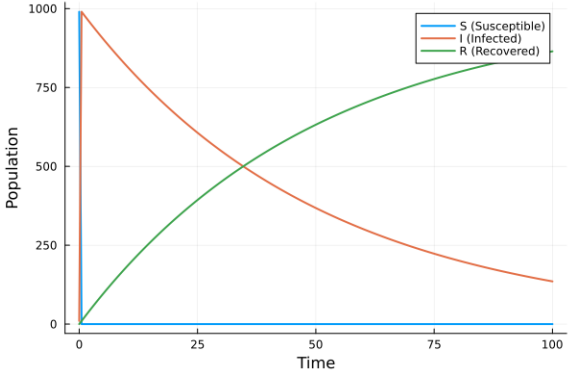
Детерминированная симуляция

```
]: for beta in β
       for gamma in γ
           df_det = simulate_deterministic(net, u0, (0.0, tmax), saveat = 0.5, rates = [beta, gamma])
           CSV.write(datadir("sir_det.csv $beta, $gamma"), df_det)
           println("simulation for $beta, $gamma")
       end
   end
```

```
simulation for 0.9, 0.6
simulation for 0.9, 0.3
simulation for 0.9, 0.02
simulation for 1.0, 0.6
simulation for 1.0, 0.3
simulation for 1.0, 0.02
simulation for 0.75, 0.6
simulation for 0.75, 0.3
simulation for 0.75, 0.02
```

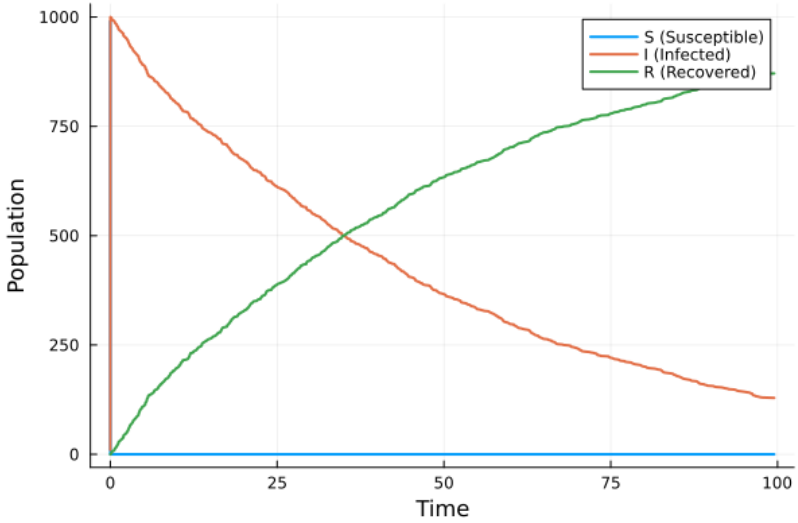
5:

5.7


S (Susceptible)
I (Infected)
R (Recovered)
Population
Time
1000
750
500
250
0
0
25
50
75
100

6: SIR

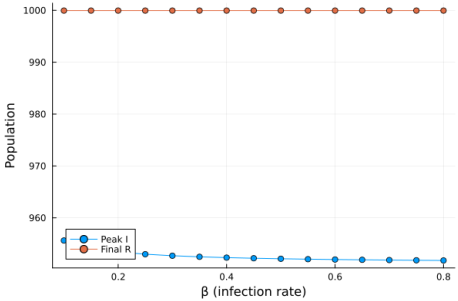
5.8


S (Susceptible)
I (Infected)
R (Recovered)
Population
Time
1000
750
500
250
0
0
25
50
75
100

5.9 β

β. ,

, (. 8) CSV .



8: β

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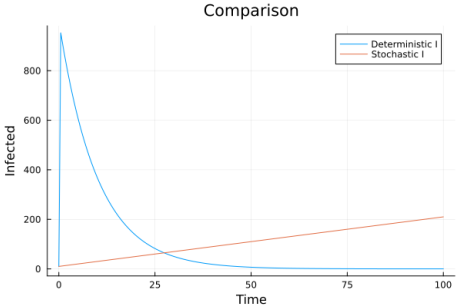
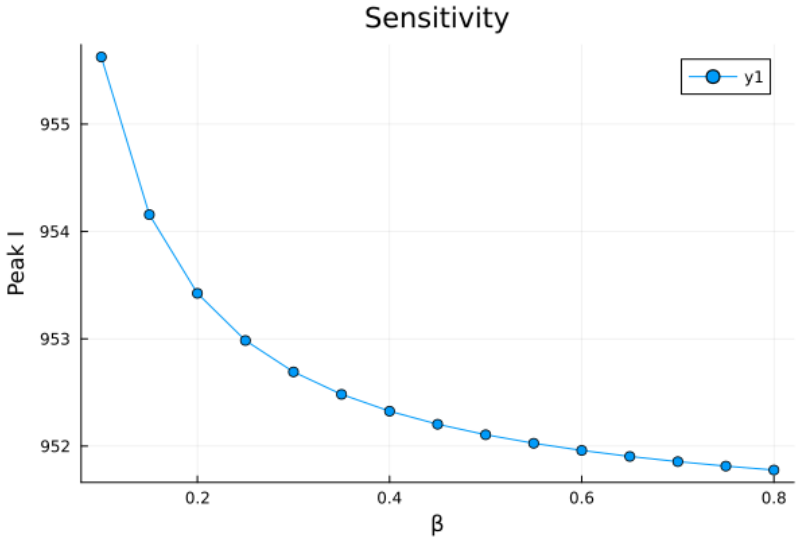

Comparison
Deterministic I
Stochastic I
800
600
400
200
0
Infected
0
25
50
75
100
Time

9:

5.12



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6.

6.

SIR .